THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The LCM of 17 and 19 is: 1

(a) 1 **(b)** 36 **(c)** 17 **(d)** 323

2. The product of the HCF and LCM of 12 and 18 is: 1

(a) 36 **(b)** 216 **(c)** 108 **(d)** 6

3. The graph of $y = p(x)$ crosses the x-axis at exactly 4 points. The number of zeroes of $p(x)$ is: 1

(a) 3 **(b)** 0 **(c)** 4 **(d)** 2

4. The zeroes of $x^3 - 4x$ are: 1

(a) $0, 4$ **(b)** $0, 2, -2$ **(c)** $2, -2$ **(d)** $4, -4$

5. The lines $2x - y = 1$ and $4x - 2y = 2$ are: 1

(a) parallel **(b)** perpendicular **(c)** coincident **(d)** intersecting

6. The line $2x + y = 6$ meets the y-axis at: Student A says $(0, 6)$; Student B says $(3, 0)$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

7. A root of $2x^2 + x - 6 = 0$ is: 1

(a) 3 **(b)** $-\frac{3}{2}$ **(c)** $\frac{3}{2}$ **(d)** 2

8. A student states: “The only consecutive integers whose product is 30 are 5 and 6.” The correct response is: 1

(a) Correct **(b)** Incorrect — -6 and -5 also work

(c) Only $-5, -6$ work **(d)** None exist

9. If $a_n = 2n + 1$, the first three terms are: 1

(a) $3, 4, 5$ **(b)** $2, 4, 6$ **(c)** $1, 3, 5$ **(d)** $3, 5, 7$

10. The number of terms in the AP $10, 13, \ldots, 100$ is: 1

(a) 29 **(b)** 32 **(c)** 31 **(d)** 30

11. In $\triangle ABC$, $DE \parallel BC$, $\frac{AE}{EC} = \frac{1}{2}$ and $AD = 3$ cm. Then DB is: 1

(a) 6 cm **(b)** 1.5 cm **(c)** 3 cm **(d)** 9 cm

12. SAS similarity requires: 1

(a) one angle equal and the sides including it proportional
(b) a right angle
(c) two angles equal
(d) all sides proportional

13. If $\cos A = \frac{3}{5}$, then $\sec A$ is: 1

(a) $\frac{5}{3}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{5}{4}$ **(d)** $\frac{4}{5}$

14. Tangents TP and TQ are drawn to a circle with centre O. If $\angle PTQ = 50^\circ$, then $\angle POQ$ is: 1

(a) 130° **(b)** 40° **(c)** 50° **(d)** 100°

15. A student states: "A sector of angle 180° is a semicircle." The correct response is: 1

(a) Correct **(b)** Incorrect
(c) It is a quadrant **(d)** Cannot be determined

16. The area of a quadrant of a circle of radius 21 cm is: 1

(a) 693 cm² **(b)** 1386 cm² **(c)** 173.25 cm² **(d)** 346.5 cm²

17. Two solid hemispheres of radius r are joined base to base. The surface area of the solid is: 1

(a) $3\pi r^2$ **(b)** $6\pi r^2$ **(c)** $2\pi r^2$ **(d)** $4\pi r^2$

18. A student states: "The modal class is the class with the highest class mark." The correct response is: 1

(a) Correct
(b) It is the last class
(c) Incorrect — it is the class with the highest frequency
(d) Cannot be determined

19. The upper limit of the class 15–25 is: 1

(a) 15 **(b)** 25 **(c)** 10 **(d)** 20

20. A coin is tossed three times. The probability of three heads is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{8}$

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Find the HCF and LCM of 24 and 36 by prime factorisation and verify that HCF × LCM = product of the numbers. 2

22. Find a quadratic polynomial whose sum and product of zeroes are 0 and $\sqrt{5}$ respectively. 2

23. Solve: $2x^2 - x + \frac{1}{8} = 0$. 2

24. $\triangle ABC \sim \triangle PQR$ with $\angle A = 32^\circ$ and $\angle R = 65^\circ$. Find $\angle B$. 2

25. Find the distance of the point $(-5, 12)$ from 2
(i) the origin
(ii) the point $(0, 12)$.

26. If $\sqrt{3}\tan\theta = 1$, find $\sin^2\theta - \cos^2\theta$. 2

27. An aeroplane at a height of 1000 m is seen at an angle of elevation of 30° from A. Find its horizontal distance from A. 2

28. A circle touches all four sides of quadrilateral ABCD with $AB = 7$ cm, $BC = 8$ cm and $CD = 9$ cm. Find AD. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $x + 2y = 4$ and $2x + 4y = 12$. 3

30. Find the median height: 3

Height (cm)	140–145	145–150	150–155	155–160	160–165
Students	3	7	10	6	4

31. An AP has 37 terms. The sum of the three middle-most terms is 225 and the sum of the last three terms is 429. Find the AP. 3

32. A circle with centre O is inscribed in quadrilateral ABCD with $\angle B = 90^\circ$. If $AD = 23$ cm, $AB = 29$ cm and $DS = 5$ cm (S the point of contact on AD), find the radius. 3

33. Find the area of the minor segment of a circle of radius 12 cm when the sector angle is 90°. (Use $\pi = 3.14$) 3

34. A jar has 54 marbles — blue, green and white. P(blue) $= \frac{1}{3}$ and P(green) $= \frac{4}{9}$. Find the number of white marbles. 3

35. From the top of a 60 m lighthouse, the angles of depression of two boats on opposite sides of it are 30° and 45°. Find the distance between the boats. 3

OR

From a point A, the angle of elevation of a tree top is 30°. On walking 30 m towards the tree, it becomes 60°. Find the height of the tree and the distance of the second point from the tree. 3

36. Find the mean by the step-deviation method: 3

Age (yrs)	25–35	35–45	45–55	55–65	65–75
Persons	6	11	7	4	2

OR

Find the mode of: 3

Class	0–10	10–20	20–30	30–40	40–50
Frequency	7	14	13	12	4

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** On a city map, a home is at H(1, 2) and a school at S(7, 10). 5

(i) Find HS.

(ii) A shop lies on HS dividing it in the ratio 2 : 3 from H. Find the shop's position.

(iii) A friend's house F is such that S is the mid-point of HF. Find F.

(iv) Find HF.

OR

Case study: In a village map, a well is at W(2, 3) and a temple at T(8, 11). 5

(i) Find WT.

(ii) A school lies on WT dividing it in the ratio 3 : 2 from W. Find its position.

(iii) A tap on the x-axis is equidistant from W and T. Find it.

38. If $x = a\sec\theta + b\tan\theta$ and $y = a\tan\theta + b\sec\theta$, prove that $x^2 - y^2 = a^2 - b^2$. 5

OR

Prove that $\frac{\tan A+\sec A-1}{\tan A-\sec A+1} = \frac{1+\sin A}{\cos A}$. 5

39. **Case study:** A test tube is a cylinder of radius 1 cm and height 10 cm with a hemispherical bottom. 5

(i) Find its capacity.

(ii) Find its outer surface area.

(iii) How many such tubes can be completely filled from 1 litre?

OR

Case study: A water bottle is a cylinder of radius 4 cm and height 15 cm with a hemispherical top. (Use $\pi = 3.14$) 5

(i) Find the volume of the cylindrical part.

(ii) Find the volume of the top.

(iii) Find the capacity in litres.

(iv) Find the curved area of the cylinder covered by a label.

40. The median of the following data is 46 and $\Sigma f = 230$. Find x and y. 5

Class	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	12	30	x	65	y	25	18

OR

The mean of the distribution is 25. Find p, and then find the median. 5

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	18	15	p	6

END OF QUESTION PAPER **Total: 80 Marks**

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